Computational Control Certificate

CTRL-04 — Terminal characterization

Erdős #81, Paper II — internal validation evidence

Reference. Paper II, Lemma 5.1
Date. 2026-07-07 **Verdict.** **PASS**

Statement audited

A finite chordal graph in which every two *nonadjacent simplicial* vertices have equal open neighborhoods is complete-split ($K_p \vee \overline{K_q}$).

Method

Exhaustive enumeration of *all* labelled graphs on $1 \leq n \leq 6$ vertices; each is tested for chordality (MCS/PEO), the hypothesis (H) on nonadjacent simplicial pairs is checked, and every graph satisfying (H) is tested for the complete-split property. Purely combinatorial (no LP).

Verbatim output

```
Exhaustive over ALL labelled graphs on n=1..6 vertices
graphs enumerated             : 33867
chordal graphs                : 3271
chordal & satisfy hypothesis(H): 25
of those, NOT complete-split   : 0  (violations of Lemma 5.1)
RESULT: PASS
```

Reproducibility

The exact script `control.py` and its verbatim output `results.txt` are included in this folder. Rerunning `python3 control.py` reproduces `results.txt`. This certificate reports a computational check on finite instances; it is supporting evidence, not a substitute for the analytic proof in Paper II. File integrity is recorded in `SHA256SUMS.txt` (this folder) and in the master `SHA256SUMS.txt` of the package.